

OLTP vs. OLAP (using MySQL and TPC-C)

Data Intelligence and Learning ([DIAL](#)) Lab

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Concepts of OLTP and OLAP

What is MySQL?



- It is the world's most popular open-source relational database management system (RDBMS).

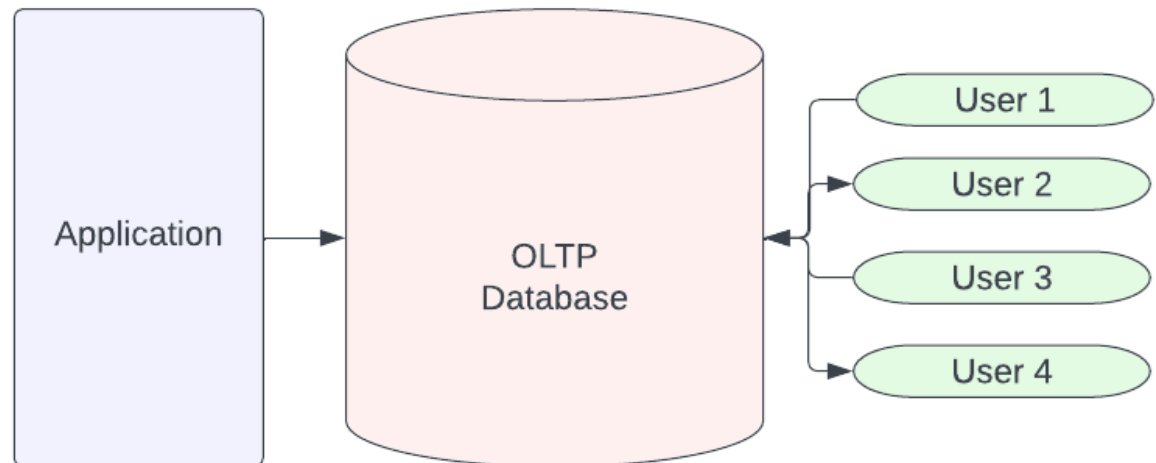


- It is fast, reliable, scalable, and easy to use.
- It was originally developed to handle large databases and has been used in highly demanding production environments.
 - ◆ It is optimized for **On-Line Transaction Processing (OLTP) workload**.

Online Transaction Processing (OLTP)



- It is a type of database system used in **transaction-oriented applications**, such as many operational systems.
 - ◆ **Online** refers to that such systems are expected to respond to user requests and process them in **real-time process transactions**.
- **The data access patterns of applications are often row-driven.**
 - ◆ Examples: ATM machines, online banking/booking/shopping
 - When we create a user, we create a new row.
 - When we retrieve a user, we get one particular row.

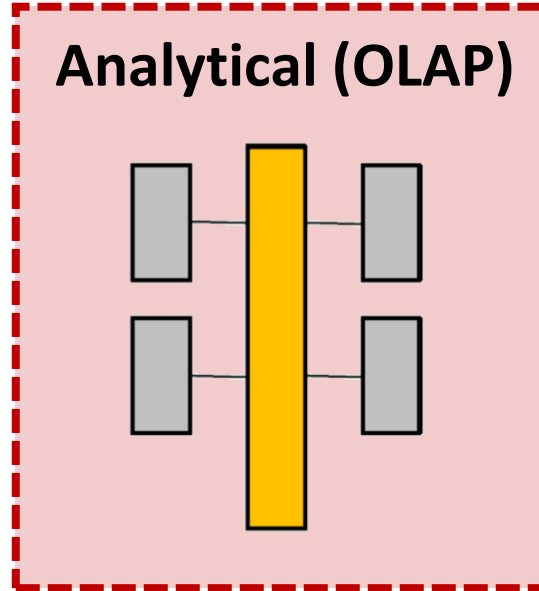


Before NoSQL



Relational

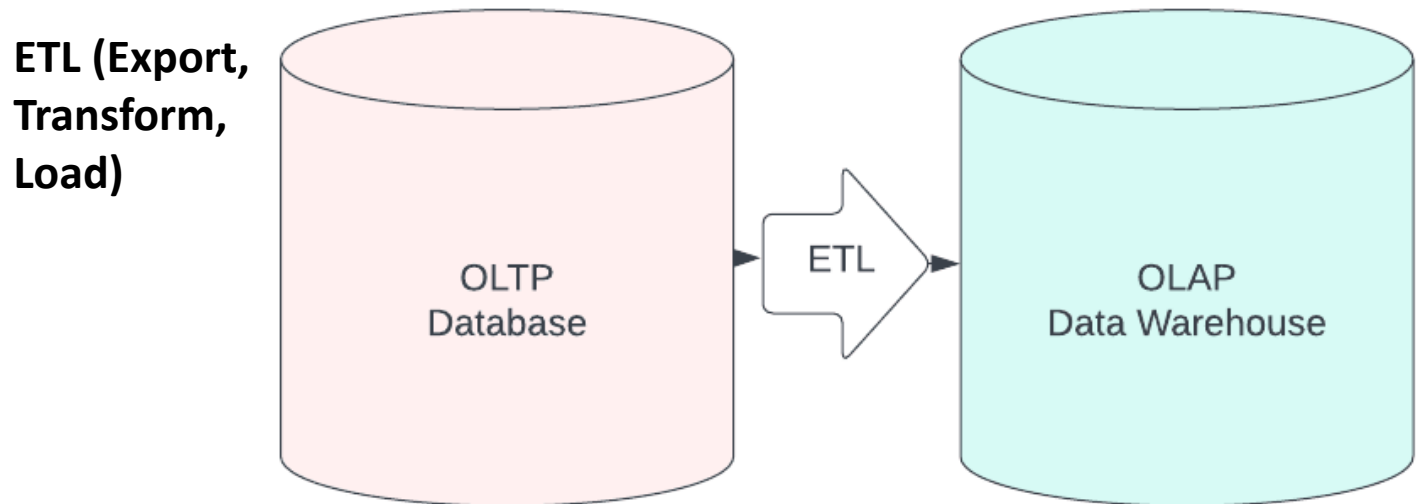
Analytical (OLAP)



On-Line Analytical Processing (OLAP)



- It is an **analytical method** that enables users to easily and selectively extract and query data from different views.
- It requires **heavy read workloads** (for large data).
- Examples: Sales analysis, market research, forecasting



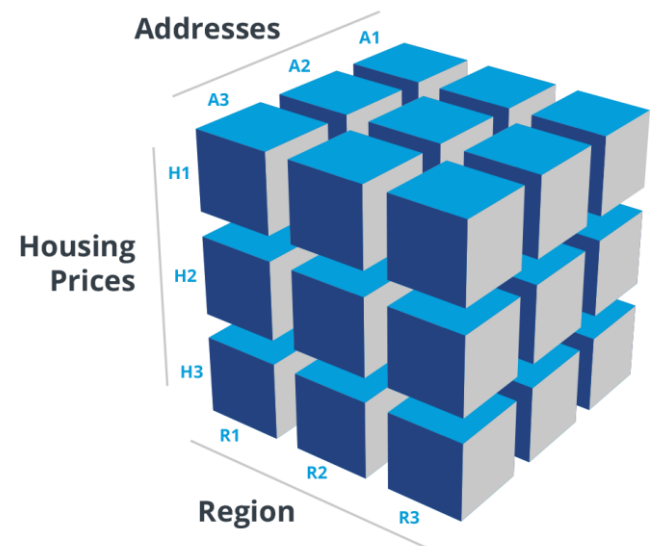
Why OLAP?



➤ For business users, it enables fast and intuitive access to centralized data and related calculations.

➤ It offers five key benefits.

- ◆ Business-focused multidimensional data
- ◆ Business-focused calculations
- ◆ Trustworthy data and calculations
- ◆ Speed-of-thought analysis
- ◆ Flexible, self-service reporting



OLTP vs. OLAP



Online Transaction Processing (OLTP)	On-Line Analytical Processing (OLAP)
Transactional Fast processing Normalized Current data 	Analytical Show queries Denormalized Historical data 
Business data warehouse	Business processing

OLTP vs. OLAP: Access Patterns

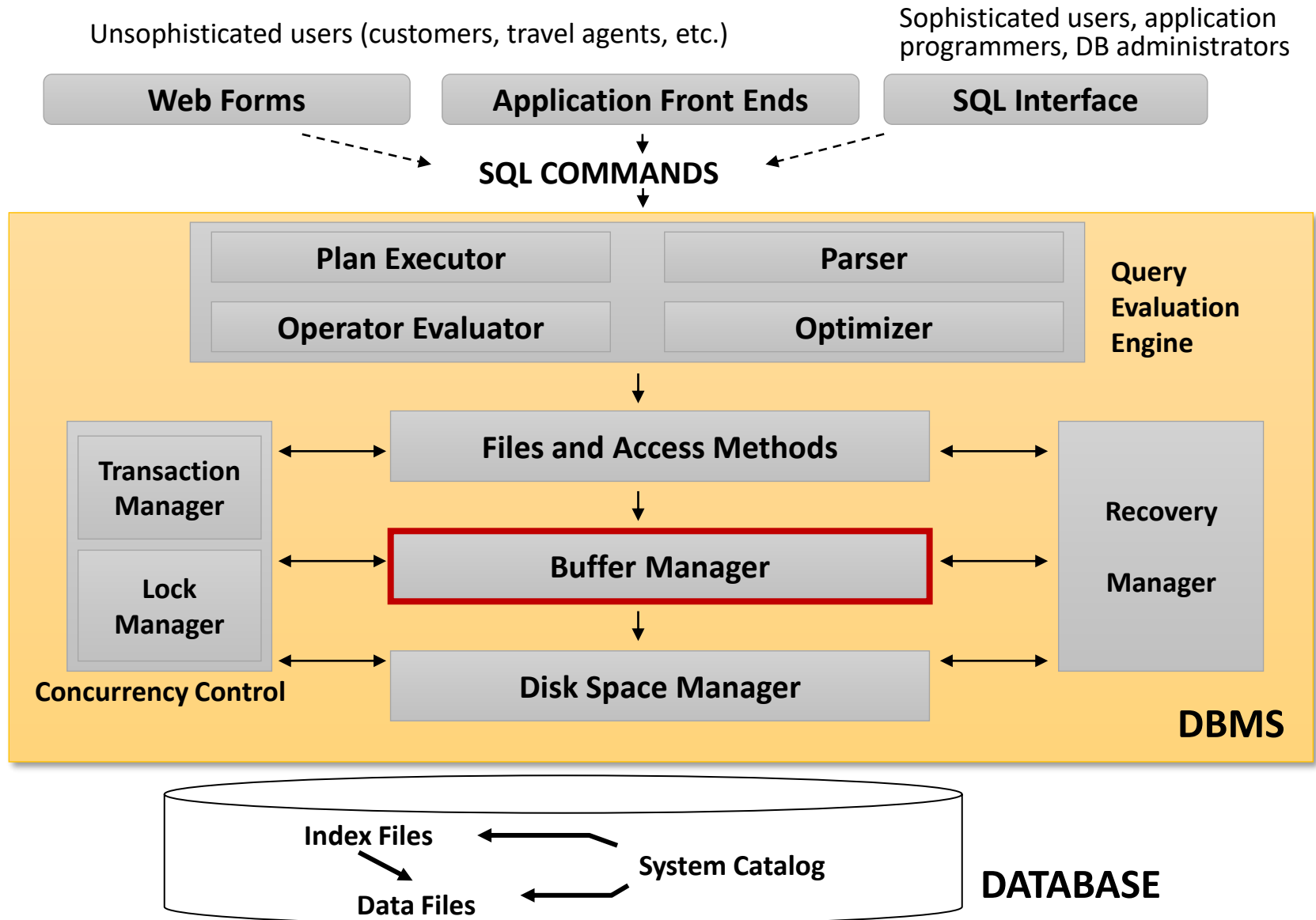


- Both OLTP and OLAP use indexes, but they access and handle data differently.

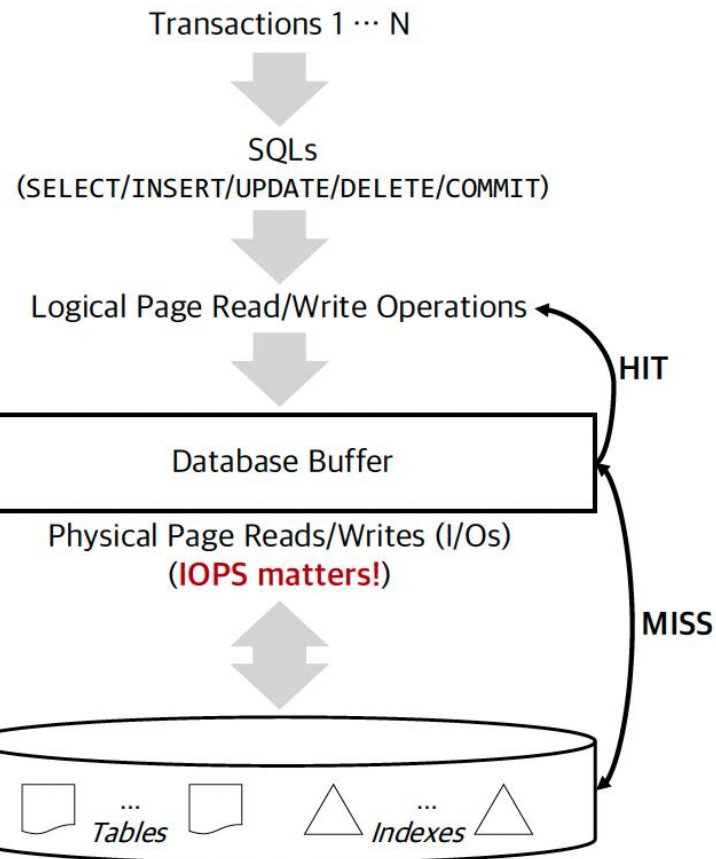
- OLTP is a **random read/write workload**.
 - ◆ It consists of random access with typically 2KB~16KB request sizes.
 - ◆ It uses index scans.

- OLAP is a **heavy-read workload**.
 - ◆ Typically, it consists of sequential accesses to a large amount of data.
 - ◆ It mainly uses full table scans.

Anatomy of an RDBMS



Database I/O Architecture



- **A transaction**
 - = A sequence of SQL statements
 - = A sequence of Reads and Writes
- **SELECT reads tuples from page(s)**
- **INSERT/DELETE/UPDATE change records in page(s)**
 - ◆ Thus, they access one or more pages
- **When page(s) is in buffer (i.e., HIT): DRAM operation**
- **Otherwise (i.e., MISS): Disk I/Os**
 - ◆ In case of dirty, write the page to storage
 - ◆ Read page(s) from storage



Concept of Data Warehouses

What are Data Warehouses?



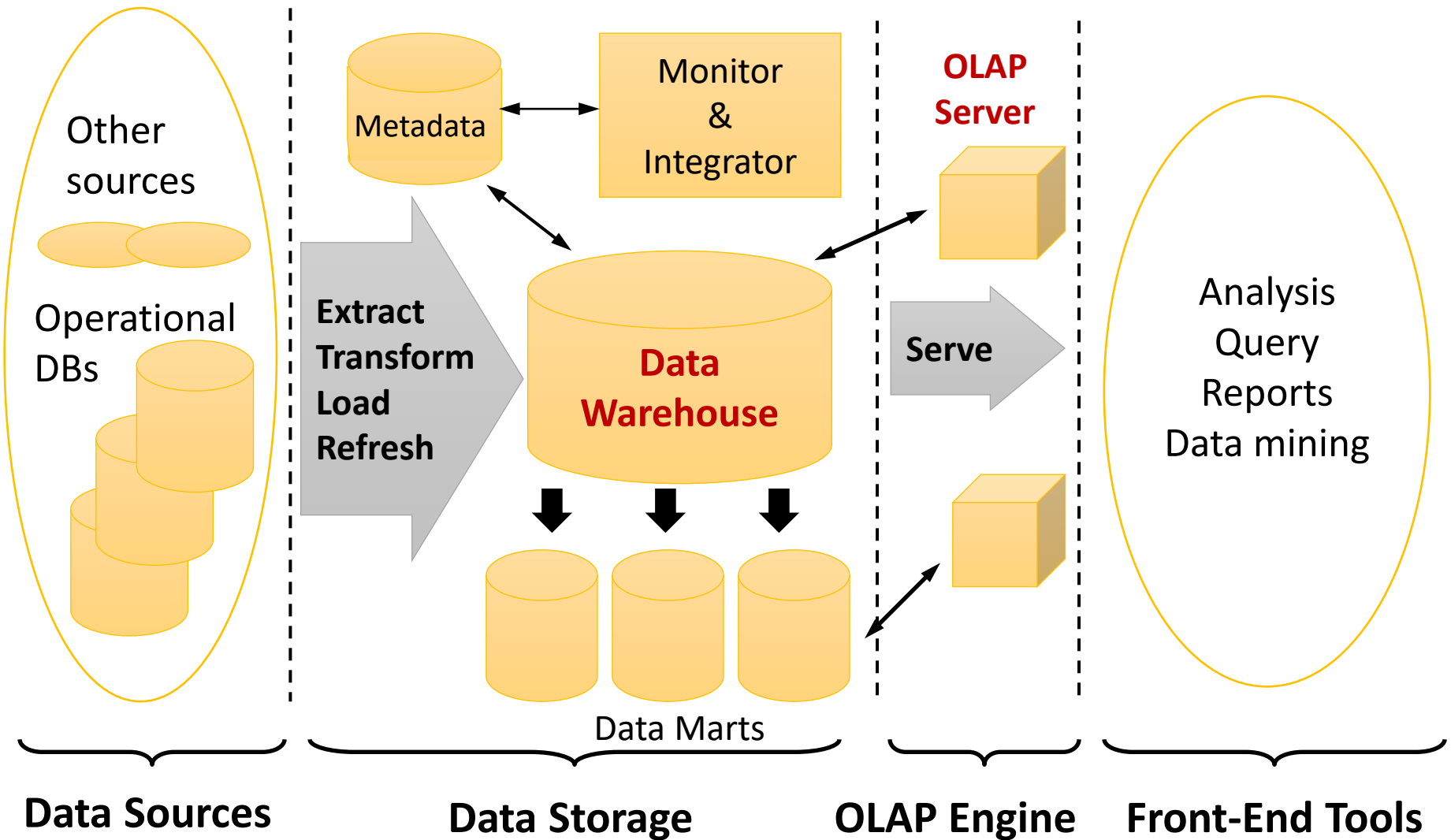
➤ It is defined in many ways, but not rigorously.

- ◆ A decision support database that is maintained **separately** from the organization's operational database
- ◆ Support **information processing** by providing a solid platform of **consolidated, historical** data for analysis.
- ◆ A data warehouse is a **subject-oriented, integrated, time-variant,** and **nonvolatile** collection of data in support of management's decision-making process.

➤ Data warehousing

- ◆ The process of constructing and using data warehouses

Architecture of Data Warehouses

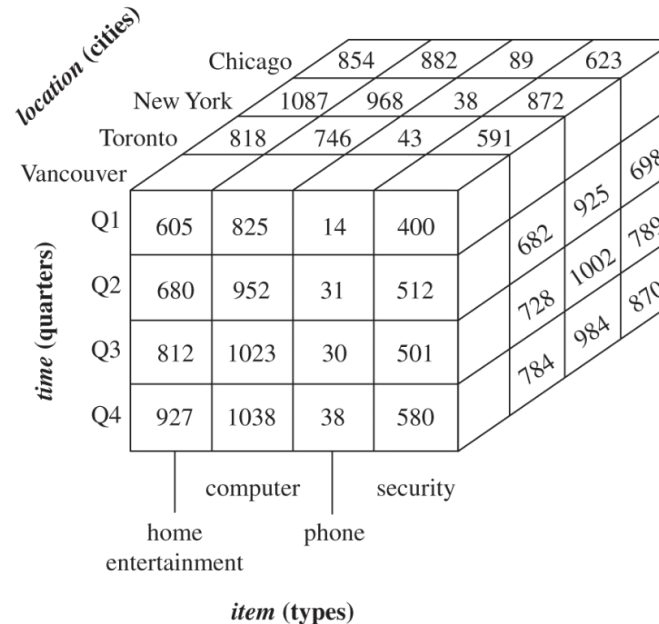


What are Data Cubes?

➤ A data warehouse is

- ◆ A **multi-dimensional data model**
- ◆ Viewing data in the form of a **data cube**

➤ Data cube can be used to form the lattice of cuboids.



Modeling of Data Warehouses



- **A data cube allows data to be modeled and viewed in multiple dimensions**
 - ◆ **Fact table** contains measures (such as dollars_sold) and keys to each of the related dimension tables
 - ◆ **Dimension tables**, such as item (item_name, brand, type), or time(day, week, month, quarter, year)

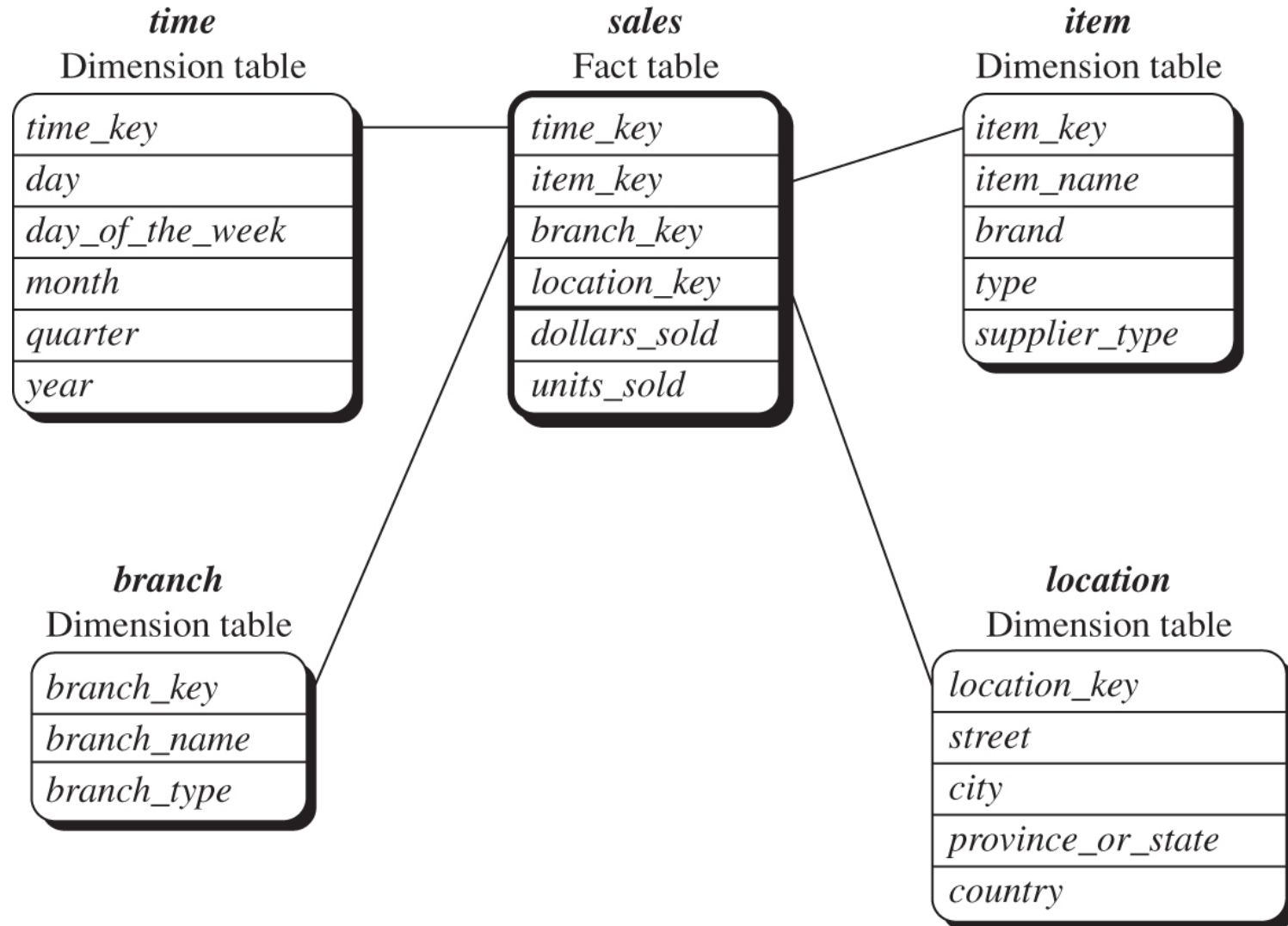
Modeling of Data Warehouses



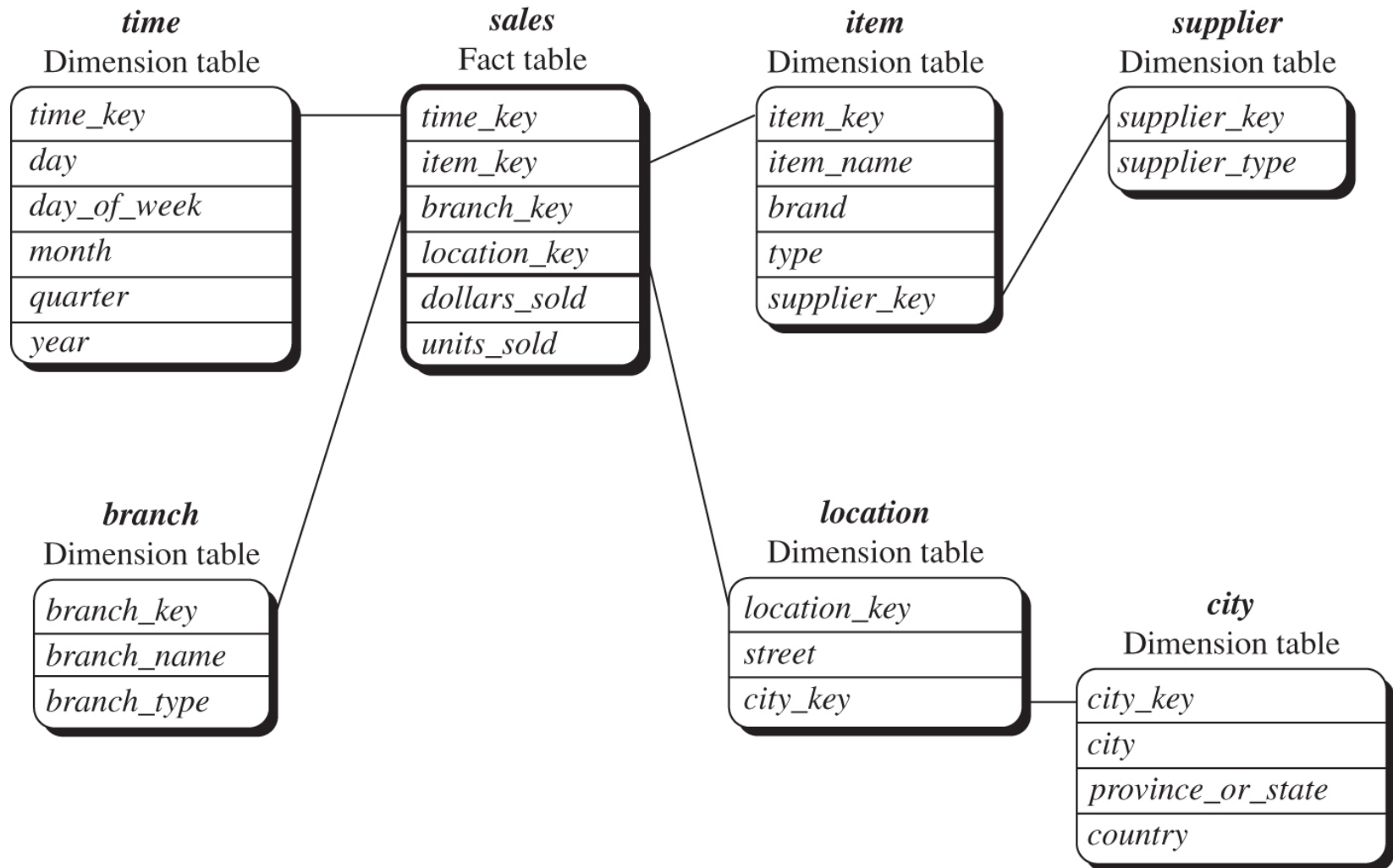
➤ Modeling data warehouses: dimensions & measures

- ◆ **Star schema:** A fact table in the middle is connected to a set of dimension tables.
- ◆ **Snowflake schema:** A refinement of star schema where **some dimensional hierarchy is normalized into a set of smaller dimension tables.**
- ◆ **Fact constellations:** **Multiple fact tables** share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation.

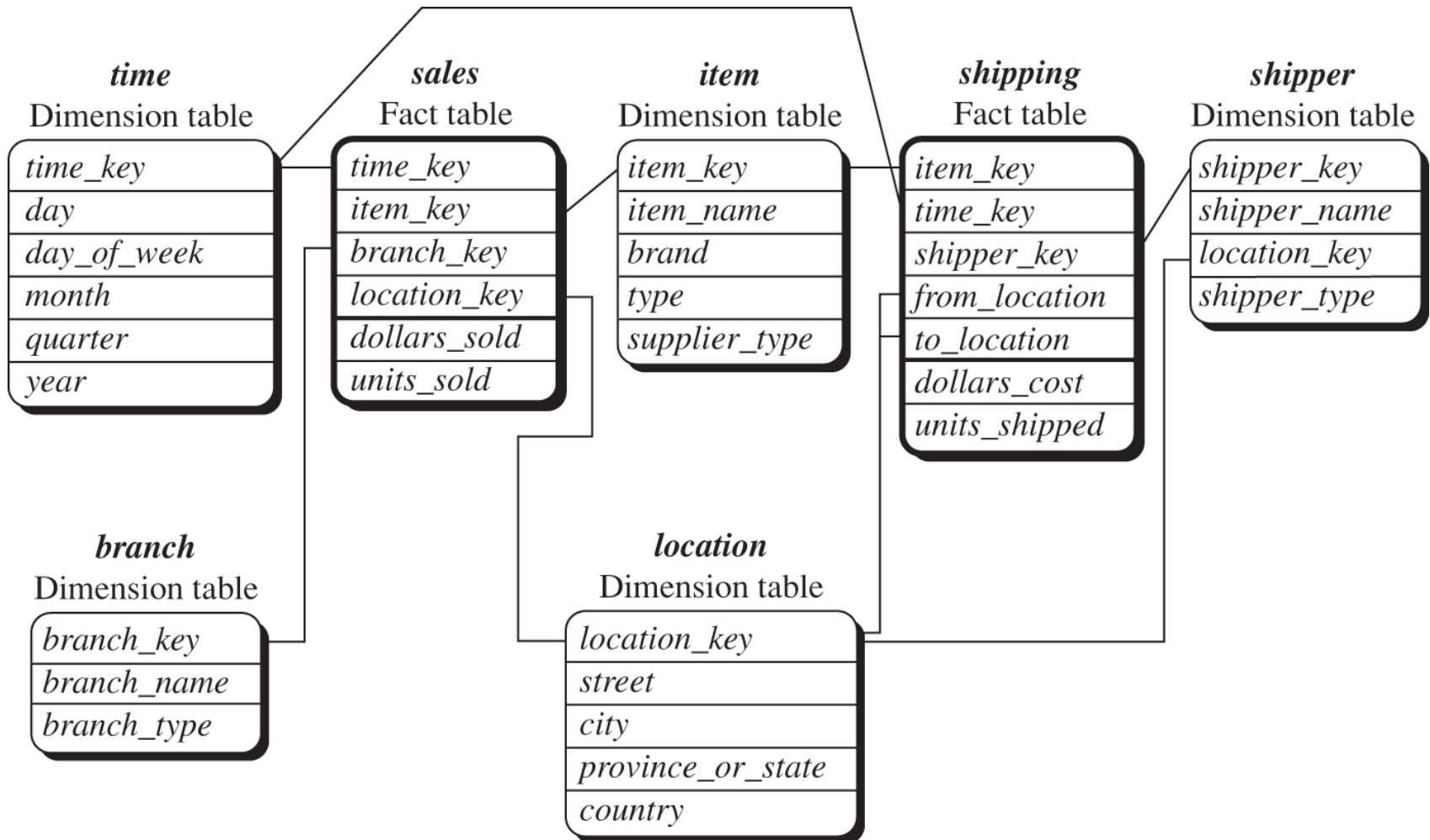
Star Schema



Snowflake Schema



Fact Constellation



How to Use Data Cubes?



➤ How to compute the total sales per item and quarter?

- ◆ Aggregate the cube by *location*.
- ◆ 3D cube is projected into 2D table.

➤ How to compute the total sales per item?

- ◆ Aggregate the cube by *location* and *time*.

➤ How to compute the total sales.

- ◆ Aggregate the cube by *location*, *item*, and *time*.

		location (cities)							
		Chicago	New York	Toronto	Vancouver	Q1	Q2	Q3	Q4
time (quarters)	computer	854	1087	818	605	825	680	812	927
	security	882	968	746	14	952	31	1023	1038
	home	89	38	43	400	512	501	580	
	phone	623	872	591	682	1002	984		
		entertainment	698	789	870				
		item (types)							

OLAP Operations



➤ Roll up (drill-up)

- ◆ Summarize data by climbing up hierarchy or by dimension reduction.

➤ Drill down (roll down)

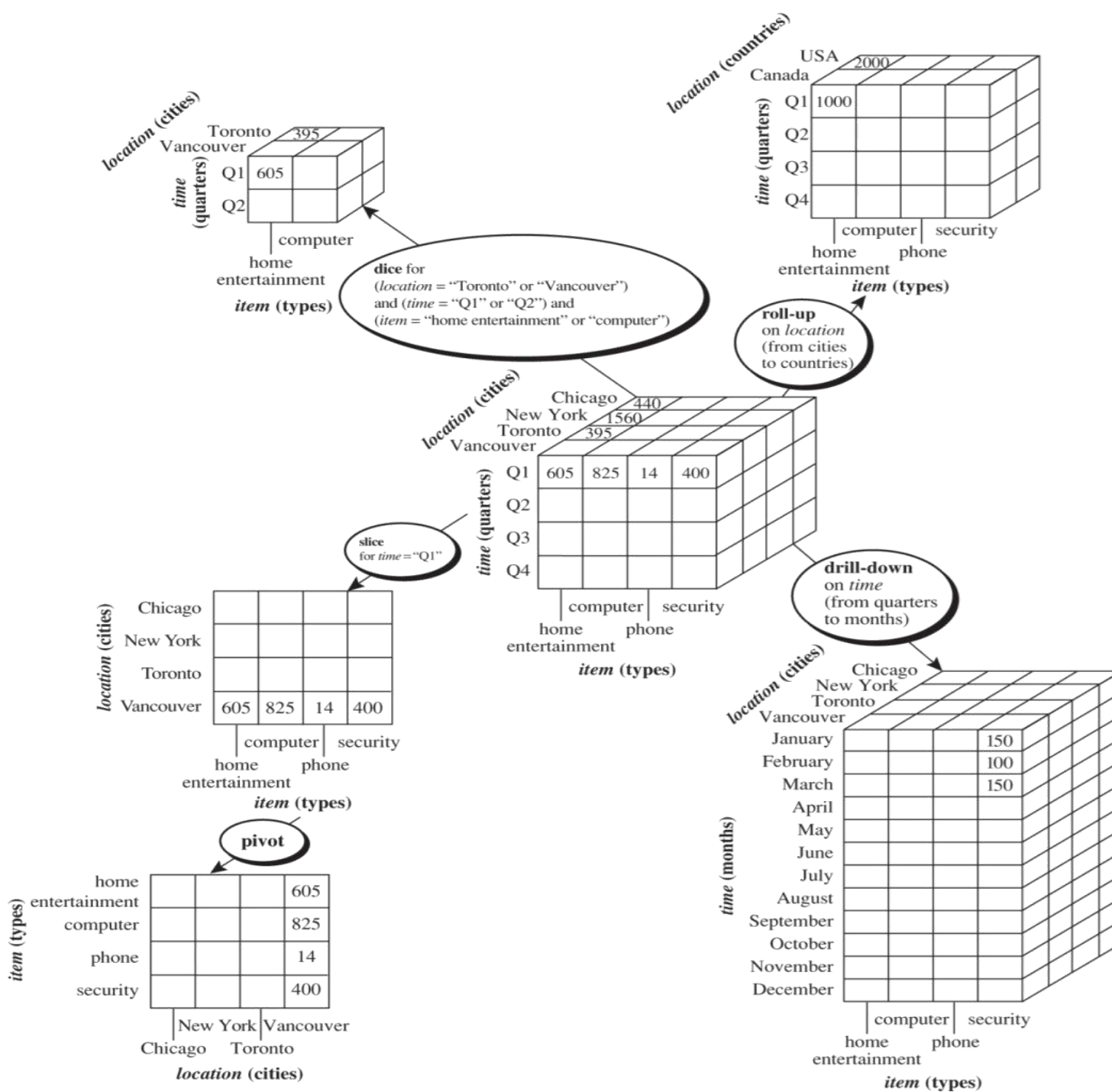
- ◆ Reverse of roll-up from higher level summary to lower-level summary or detailed data or introducing new dimensions.

➤ Slice and dice

- ◆ project and select.

➤ Pivot (rotate)

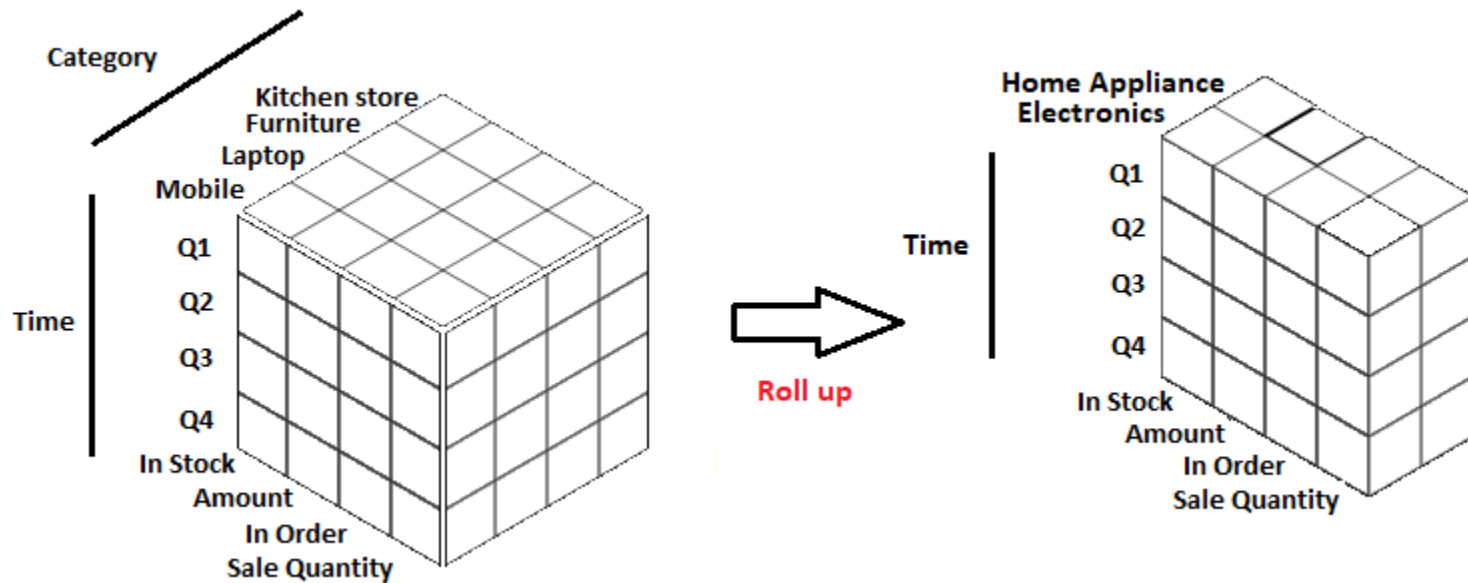
- ◆ re-orient the cube, visualization, 3D to series of 2D planes.



Roll-up Operation

➤ Decrease data granularity.

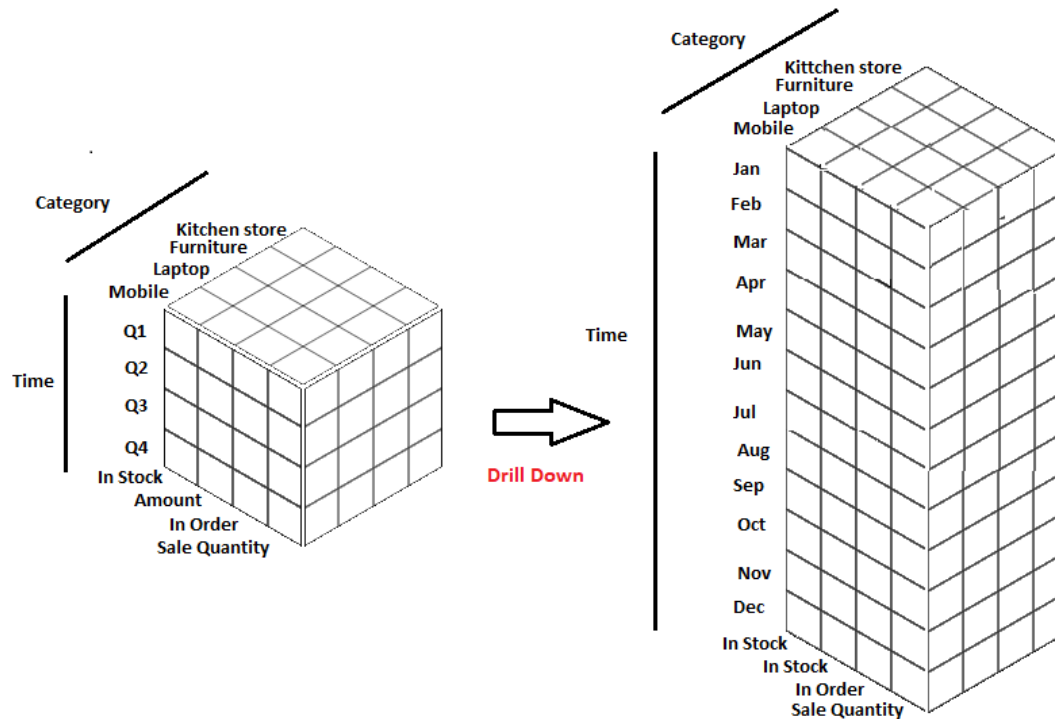
- ◆ Dimension reduction
- ◆ Perform aggregation on a data cube by climbing up a concept hierarchy for a dimension.



Drill-down Operation

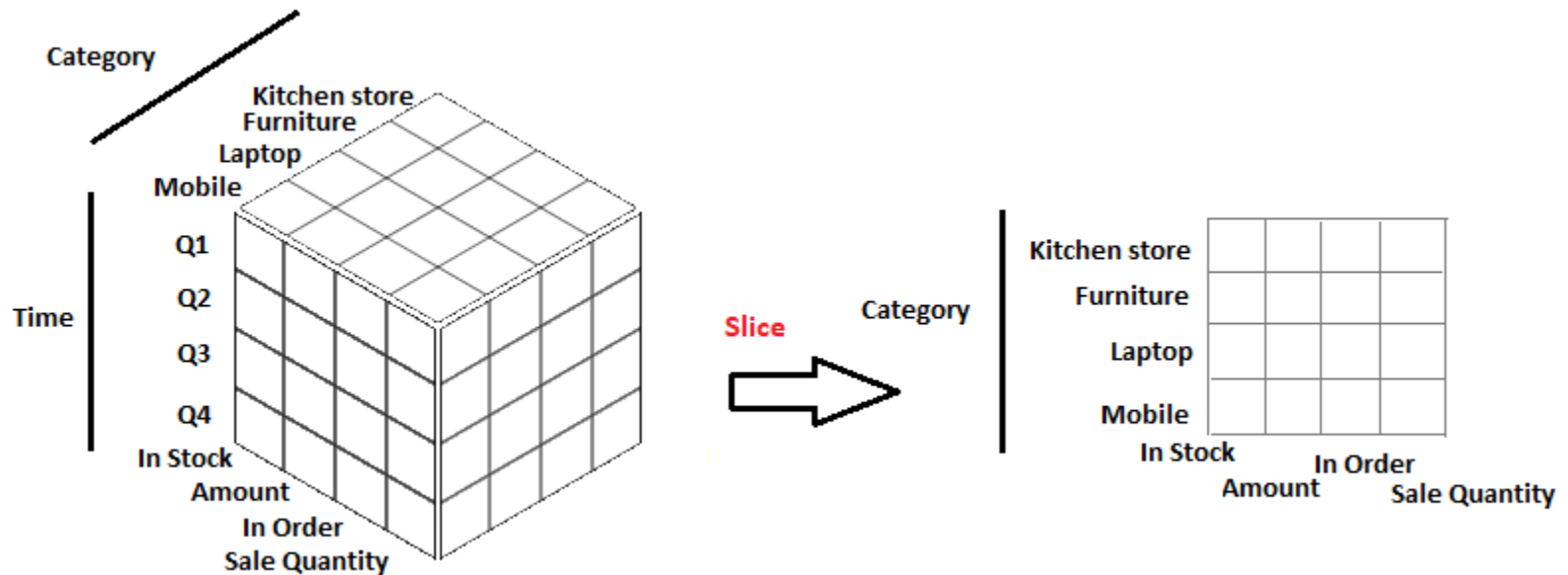
➤ Increase data granularity.

- ◆ Introducing additional dimensions.
- ◆ Navigates from less detailed data to more detailed data by stepping down a concept hierarchy for a dimension.



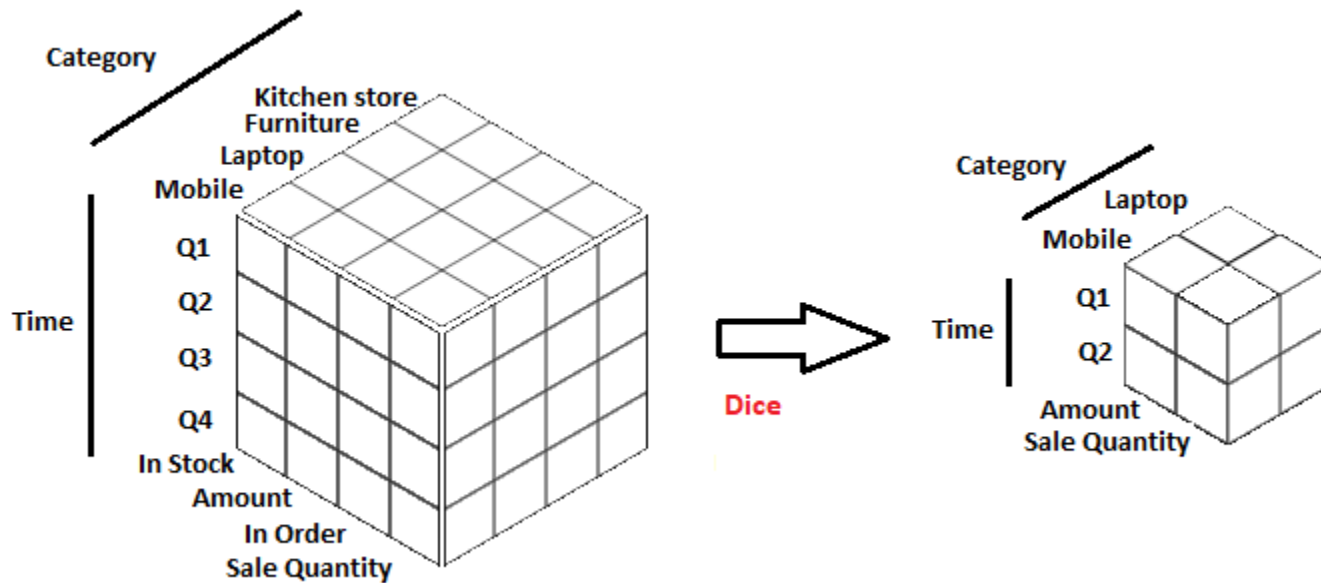
Slice Operation

- Performs a selection on one dimension of the given cube and provide a sub-cube.



Dice Operation

- Selects two or more dimensions from a given cube and provides a new sub-cube.



Q&A

